

SET ANSWER 1

Scenario based answer

Scenario 1: Identifying Trend

Logic:

- A trend means the data is generally moving upward or downward over time.
- First, plot the data against time.
- If the values consistently increase or decrease, it suggests a trend.
- We can confirm it using moving averages or trend decomposition.

Scenario 2: Detecting Seasonality

Logic:

- Seasonality means a pattern repeats at a fixed time interval.
- Here, electricity usage increases every summer.
- Plot the monthly data and look for repeated yearly patterns.
- We can also use seasonal decomposition or ACF to confirm it.

Scenario 3: Making Data Stationary for ARIMA

Logic:

ARIMA generally works better with a **stationary time series**.

If the data has:

- Trend → use differencing
- Seasonality → use seasonal differencing
- Non-constant variance → consider a transformation such as log transformation

Scenario 4: Choosing Model for No Seasonality

The data has:

- No seasonality
- Clear trend

A suitable choice is Holt's Linear Trend method

Logic:

- It handles a changing level + trend.
- Since there is no seasonality, we don't need a seasonal model.

Scenario 5: Evaluating Forecast Performance

Logic:

1. Split the historical data into **training and testing data**.
2. Train the forecasting model using the training data.
3. Forecast the test period.
4. Compare predicted values with actual values.
5. Calculate evaluation metrics such as:
 - MAE
 - RMSE
 - MAPE
6. Lower error generally means better forecasting performance.

Scenario 6: Sudden Spike in Data

Logic:

A sudden high value could be:

- A genuine event
- A data-entry error
- An outlier

Scenario 7: Sales Goes Up Every December

This indicates **seasonality**.

Logic:

- December is a fixed period.
- The increase happens repeatedly every year.
- Therefore, there is a yearly seasonal pattern.

Scenario 8: Forecasting with Missing Values

For stock prices, if a few days are missing:

Logic:

1. Identify the missing dates.
2. Check why the values are missing.
3. If those dates are **non-trading days** such as weekends/market holidays, don't fill them.
4. If actual trading-day data is missing, use an appropriate method such as **forward fill**, interpolation, or another suitable imputation method, depending on the situation.
5. Make sure the time series has the correct chronological order before forecasting.